

2) What is multiple linear regression? state how to use multiple linear regression with its relevant formula.

→ Multiple linear regression is a statistical technique used to model relationship between one dependent variable and two or more independent variables. Unlike simple linear regression which deals with only one independent variable, multiple linear regression allows us to analyze how several factors simultaneously influence the dependent variable.

This method is particularly useful when you want to predict an outcome based on the multiple predictors. It provides a way to quantify the effect of each predictor while controlling for the influence of the others.

Formula for multiple linear regression

General formula for multiple linear regression is

$$Y = a + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

where

- Y is the dependent variable (outcome you are trying to predict)

- x_1, x_2 are independent variables (the predictors)

- a is the intercept (the predicted value of Y when all x are zero)

- $b_1, b_2 \dots b_n$ are the coefficients (slopes) that represents the change in Y for a one-unit change in each corresponding x .

How to use multiple linear regression

1. Collect and prepare data

- Gather the data for dependent Variable and all relevant independent variables

For example: If you want to predict house price, your dataset might include variables like square foot, number of bedrooms, location and age of the house

2. Check assumptions

- linearity: The relationship between the dependent Variable and each independent Variable
- Independence: The observations should be independent of each other
- Homoscedasticity: The variance of errors should be consistent across all levels of the independent variables
- Normality: The residuals (errors) should be normally distributed

3. Fit the model

- Use Statistical Software to estimate the Coefficients $b_1, b_2, \dots, b_n$ and intercept a
- The Software will provide regression equation, which can be used to make predictions

4. Interpret the result

- The intercept a indicates the predicted value of Y when all independent variables are zero
- Each coefficient b_i shows how much dependent variable Y changes for one-unit increase in corresponding independent variable x_i , holding all other variables constant

5. Make predictions

- Use the regression equation to predict dependent Variable Y for new values of independent Variables $X_1, X_2, \dots, X_n$

Least Square Regression

Our aim is to calculate the values m (slope) and b (y-intercept) in the equation of the line

$$y = mx + c$$

y = Dependent Variable

x = Independent Variable

m = slope or Gradient

c = y-intercept

To find the line of best fit for N points

Step 1: For each (x, y) point Calculate x^2 and xy .

Step 2: Sum all x, y, x^2 and xy which gives us $\Sigma x, \Sigma y, \Sigma x^2$ and Σxy

Step 3: Calculate Slope $m =$

$$m = \frac{N \Sigma(xy) - \Sigma x \Sigma y}{N \Sigma(x^2) - (\Sigma x)^2}$$

(N is the number of points)